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AI nudging and decision quality: Evidence from randomized experiments in online recommendation setting

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ABSTRACT

This study explores the impacts of AI nudging on customer purchase decisions. Digital nudging is a well-established technique used to alter people's behaviors in a predictable way. With the rapid development of Artificial Intelligence/Machine Learning (AI/ML) and the widespread integration of the "black box" algorithm in the digital choice architecture, personalized targeting nudges can vastly influence individual and collective behaviors and lead to undesired consequences. AI nudge refers to the situation when human outsources developing and implementing nudges to AI/ML systems. Drawing upon the literature on nudge and recommendation agents/systems in IS, this study investigated the impact of two types of recommendation badges on user decision quality: AI nudge (e.g., *Amazon's Choice*) and non-AI nudge (e.g., *Best Seller*). We found that these two badges can lead to different user perceptions of transparency and thus affect the choice confidence of product selection. In addition, the effect of perceived transparency on choice confidence is contingent upon the mismatch/match between the recommendation and users' preferences, with perceived transparency exerting significantly higher influence on choice confidence in the preference match condition. We tested our research model using a randomized experiment and post-task survey data collected from 837 US-based college students with online shopping experience. This is the first empirical study examining the impact of AI nudging on user decision-making on e-commerce platforms and will contribute to the nudge literature and biased recommendation research in IS. The study also brings ethical implications to the use of AI/ML models and calls for careful oversight on delegating the power of nudging to AI in guiding online user behavior.

1. Introduction

In 2015, Amazon introduced "*Amazon's Choice*", an AI-powered recommendation feature integrated into its Echo AI assistant. Originally designed to streamline and automate the product purchase processes of a small selection of frequently needed products, this feature quickly expanded to encompass all product categories. When a customer searches for a product with descriptive keywords, the platform automatically picks one or two products with Amazon's personalized recommendation algorithm and labels it/them in the search result list with the badge, *Amazon's Choice*. Such a recommendation intends to ease users' information overload caused by the significant amount of product-related information in the digital marketplace and nudge consumers to make fast, satisfying, and additional purchases [1]. As a result, the e-commerce platform can enhance user satisfaction, increase customer retention, and achieve higher average order value [2,3,4,5].

However, public concerns about the algorithmic rationale of *Amazon's Choice* rose as articles[12] pointed out that it does not necessarily guarantee good quality and may lead to complex situations where the mechanism is not transparent and accountable for bad

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[1] *How is 'Amazon's Choice'chosen, Amazon Won't Say* by Carnoy published on cnet.com in 2018. See https://www.cnet.com/tech/tech-industry/do-humans-choose-what-products-get-amazons-choice (last accessed Apr 17 2023).

[2] *'Amazon's Choice'Does Not Necessarily Mean A Product Is Good* published by Nguyen on BuzzFeed News in 2020. See https://www.buzzfeednews.com/article/nicolenguyen/amazons-choice-bad-products (last accessed Apr 172,023)

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recommendations. These influential online media not only warned customers not to take the face value of *Amazon's Choice* but also dug out evidence showing manipulation of product choice with opaque AI-algorithmic models as one study[3] trained a machine learning model using the random forest algorithm and the result claimed that knowing only whether a product was an Amazon brand or not can predict if the product got the top spot 70 % percent of the time, which outperforms other factors like product star ratings and review quantity. Customers have no idea how the choice is picked out from thousands of products with the same functions, shapes, and price. As online recommendations and internet searches are never neutral [6,7], personalized algorithmic recommendation may pick out products that do not match the consumers' purchase requirements or benefits. The question that how consumers will make decisions without knowing the algorithmic mechanism of the badge remains unanswered.

Most e-commerce platforms have since followed Amazon's early forays, to rely heavily on personalized algorithms to generate recommendations to target potential customers using badges. These kinds of recommendations are built on AI (artificial intelligence) and ML (machine learning) algorithms, whose predictive capability is predicated on opaque, black box processes [8,9]. They are ubiquitous across the internet, appearing through badges like Amazon's Choice on Amazon.com, OUR PICKS FOR YOU on Dsw.com, Flash pick of the day on Walmart.com, and Best Delivery on Yelp.com, among others. While these badges have reduced the friction in e-commerce transactions with more efficiency and accuracy, sometimes they cannot meet the targeted person's actual needs and benefits, causing unintended effects during consumer's decision-making process. For example, customers can be affected when trying to approach the optimal decisions due to a lack of explicit explanation of the factors leading to the recommendation result [10]. Consumers' preferences are also at risk of being manipulated unconsciously by online recommendations [11].

Compared to *Amazon's Choice*, which is driven by non-transparent AI algorithms, *Best Seller,* has a clear clarification of how it is being generated and also appears commonly on the search result page. Both recommendation badges, *Amazon's Choice* and *Best Seller*, are text-based digital nudges that manipulate the choice architecture in the online shopping context [12]. Recently, nudges in the digital environment have become a promising area of research in information systems (IS) and have received increasing attention from scholars. Digital nudges refer to the nudging technique employing user interface (UI) design elements to guide people's behaviors in digital choice environments and improve their decision-making [13]. Theoretically, drawing on dual-system theory, digital nudging affects human's mental model by targeting either System 1 or System 2-the two cognitive modes through which people process choices. The nudging technique leverages human's proneness to make decisions with minimal effort, subconsciously relying on System 1, and can establish heuristics-mental shortcuts in cognitive processing. As a result, digital nudges can affect user behavior in certain directions while retaining the freedom of choice via heuristics and cognitive biases [14]. Empirically, it is also demonstrated that users' pre-existing preferences can be manipulated by online recommendations subconsciously via cognitive biases (e.g., anchoring effects) during the product selection process [11].

There are ethical criticisms of nudging and a debate about whether to nudge. For example, Hausman and Welch [15] argue that human decisions can be affected by failed design of choice architecture on purpose. They also state that choice architecture performs best when humans are unconscious of the influence of nudges on their behaviors, which infers that the choice-retaining and non-intrusive credentials of nudges have been exaggerated. Regarding this problem, we hold the same opinion as Selinger and Whyte [16] that the social anxieties related to nudges may be inflated, but we should take the ethical concerns seriously. Moreover, we argue that before modifying anyone's behavior, we should satisfactorily determine the mechanism and evaluate both the desired and undesired outcomes of the intentional choice architecture. As the design of the choice architecture of many digital nudges has been outsourced to AI/ML algorithms, the black box characteristics of AI are also being embedded into the UI elements in all kinds of applications, platforms, and web-based services. However, how the black box characteristics will affect users' cognitive modes during decision-making remains underexplored. For example, could the opaque nature of AI-driven nudging lead to unexpected consequences within traditionally studied choice architectures? According to the current literature, the potential large-scale impacts of AI-driven digital nudges on users' collective behavior remain unanswered. Therefore, this study focuses on the recommendation badge driven by AI-algorithm on e-commerce platform and sheds light on the impact of its black box characteristic on decision quality because there is a lack of knowledge on how such AI-algorithmic recommendation badges will impact consumers' perceptions and potentially alter the public's behavior [17].

There is a significant difference between *Amazon's Choice* and *Best Seller* based on how people may perceive them. The *Amazon's Choice* badge is generated and placed next to products by an AI algorithm where Amazon states it tries to emphasize highly rated and well-priced products available to ship immediately.[4] However, consumers do not know all the features driving the placement of badge nor the subject and objective factors that contribute to ratings and quality. Such badge is also referred to as AI nudge as its deployment is delegated to AI algorithms and its placement remains largely unpredictable [5]. Compared to that, the *Best Seller* badge signifies the product with the highest sales amount in its corresponding product category, which the consumer can follow up and trace the information by clicking on the corresponding page. In this study, we conceptualize the recommendation badges with undisclosed generation factors as AI nudges (e.g., *Amazon's Choice*), and those with fully transparent generation mechanism as non-AI nudges (e.g., *Best Seller*).

As people gradually realize that most information technology is not neutral [18], empirical research is needed to show how customers are affected by the AI nudge in different contexts. In this study, we explore the influence of transparency of recommendation badges (AI vs. non-AI nudge) on user' subjective decision quality in two different situations (the recommendation matches the user's preference or not). Figuring out this question can be meaningful to the literature on nudging, biased recommendation agents/systems research, Explainable AI. Practically, this study also contributes to the ethical use of AI nudging and AI/ML-algorithmic components in the UI/UX design since many companies use the technologies but remain conservative in disclosing the algorithm details.

2. Related literature

In this section, we first review the related literature on recommendation agents in information systems, and theories of dual-system and cognitive bias in cognitive psychology. We then discuss the research in nudging and explainable AI to clarify how digital nudging has been theorized and applied in information systems research and why exploring the role of transparency in AI nudging is imperative.

[3] *How We Analyzed Amazon's Treatment of Its "Brands" in Search Results* by Leon Yin and Adrianne Jeffries published on The Markup in 2021. See https://themarkup.org/amazons-advantage/2021/10/14/how-we-analyzed-amazons-treatment-of-its-brands-in-search-results (last accessed Apr 172,023).

[4] *What Does It Mean When a Product Is 'Amazon's Choice'?* by Louise Matsakis published on Wired.com in 2019. See https://www.wired.com/story/what-does-amazons-choice-mean (Last accessed Apr 19, 2023)

2.1. Recommendation agent/system

Prior research in recommendation agents/systems (RA/RS) has studied the impact of personalized online recommendations on user decisions from different aspects and recently focused intensively on biased recommendations [19]. These studies conceptualize recommendation bias and operationalize it in two different ways. One group of researchers defines recommendation bias as when RA systematically favors a subset of products over other products regardless of users' preferences. The occurrence of such biased recommendations can be attributed to two reasons: one is third-party sponsorship or advertisement; the other is the inaccuracy or malfunction of the RA/RS [20]. For instance, Wang and Wang [10] studied the biased RAs that manipulate the recommendation in order to favor sponsored products. Krishnasamy et al. [21] consider an RS biased when ads are displayed without any disclosure information (e.g., tags) and disguised as genuine recommendations. Wang et al. [7] explored biased product recommendations that do not fit users' indicated preferences in their study. Another group of scholars defines recommendation bias as arbitrary perturbation without subjective intention. For example, Adomavicius et al. [22–24] operationalized biases as randomly assigned jokes and songs' ratings in the user interface and then the experiment subjects rate the jokes and songs accompanying those ratings.

Despite the fact that recommendation bias is defined and operationalized differently, studies from both sides confirmed that biased recommendations can exert unexpected effects on consumer's perceptions and decisions. For example, Adomavicius et al. [11] demonstrated that user preferences can be manipulated by randomly generated ratings and eliminating preference uncertainty cannot weaken such unintended anchoring effect. Adomavicius et al. [23] tested different rating presentation formats and found that graphical rating display designs are more effective than numerical designs in reducing biases, which cannot fully remove the bias. Xiao and Benbasat [25] showed that consumers are susceptible to biased personalized recommendations based on results from online experiments.

Moreover, prior studies have examined customers' perceptions of personalization, trust, and integrity with biased RA and other objective factors in the recommendation environment, including explanation mechanisms, user's prior knowledge, and sponsorship disclosure [23,10,26,25]. Among these studies, perceived personalization and user trust are two antecedents of user decision quality in the biased recommendation context, where RA/RS chooses a product that does not match the user's preferences. For example, Xiao and Benbasat [25] unveiled the mediating role of perceived personalization in facilitating the negative influence of biased recommendations on users' decision quality and decision effort, where perceived personalization refers to how much consumers perceive that the website understands their preferences and needs of the corresponding product. The results showed that personalization can be used to increase the acceptance of deceptive recommendations. Wang et al. [7] found that user's trust in biased RA increases only when explanations for organic/unbiased recommendations and sponsorship disclosure are both provided. Wang and Wang [10] explored sponsored recommendations in a scenario where users assume the RA analyzes their historical preferences to generate recommendations. The findings suggested that users' perceived RA integrity is increased with sponsorship disclosure under the condition that users have high prior knowledge about the prevalence of sponsorship. They also identified perceived RA transparency as a mediator in the impact of sponsorship disclosure on perceived integrity, where they defined transparency as users' understanding of the inner workings of the RA and specifically referred to the sponsorship disclosure.

Unlike prior studies, this work explores nudging-based recommendations (i.e., presenting a recommendation badge beside an item) on e-commerce websites and expands the research of nudging recommendations in IS. Specifically, we compare users' perceptions of transparency between AI and non-AI nudge and investigate how perceived transparency affects user's decision quality in two scenarios: preference match and preference mismatch, which simulates the real-world situation on shopping websites and is the same as one of the aforementioned definitions of recommendation bias: the recommendation favors a small set of products over the others despite of user's preferences. This study uses preference mismatch rather than recommendation bias to avoid ambiguous references due to the same word used in cognitive bias theory cited in the following subsection. Moreover, this study does not dive deep into the causes of preference mismatch and only explores the differences of decision quality between preference mismatch and preference match conditions.

2.2. Dual-system theory and cognitive biases

Dual-system theory states that the human brain uses two modes of thinking: one is intuition and automatic, and the other is reflective and reasonable. These two modes are also coined as thinking *system 1* and *system 2,* respectively, in the book, *Thinking, Fast and Slow* [27]. The theory pointed out that human decision comes from either mode of thinking while both systems collaborate seamlessly in our brains. We follow the intuitive description and refer to these thought patterns as automatic and reflective thinking. Automatic thinking is the principal mode of thinking, and its most important characteristics are fast and effortless [27]. It dominates our judgment when we need to make quick decisions with minimal efforts, such as repeated and competent tasks (e.g., driving, sophisticated daily routines), which is considered as instinctive, emotional, and unconscious. Compared to that, reflective thinking is slow, effortful, rational, and based on logical reasoning [28,29]. Due to humans' predisposition to reduce effort, reflective thinking comes into action when the automatic system cannot handle it and when we need to pay attention and invoke memory resources in certain circumstances, such as solving a math equation, weighing the pros and cons of a decision, etc. It is estimated that 95 % of our daily decisions are initiated by conditional stimulus and resolved by automatic system [30]. In those situations, heuristics-mental shortcuts can enable the human's brain to substitute missing or hard-accessible information for some other readily available information, which is more likely to achieve better decision quality. Take herd instinct as an example, when we face a situation that is first encountered, we may observe what others do and simply follow the actions.

Not only can heuristics and intuition help us make efficient decisions but also make us vulnerable to cognitive biases. A cognitive bias is a systematic pattern of deviation from norm or rationality in judgment [31]. Individuals have misperceptions from the construction of reality, which may cause them to behave based on this "subjective reality". On one hand, decisions based on cognitive heuristics rely on mental shortcuts that ease the cognitive load, these mental shortcuts can smooth the process and lead to a "less is more" effect. On the other hand, these heuristics-mental shortcuts can also cause biases in certain situations [32,14].

Prior studies recorded a variety of cognitive biases, such as framing effect, status-quo bias, loss aversion, decoupling, and so on. We examined every kind of cognitive bias based on the literature summarization from Mirsch et al. [12] and teased out three types of cognitive biases that can trigger heuristics-mental shortcuts in a consumer's decision-making process with recommendation badges, which are anchoring-and-adjustment, social proof, and messenger effect. Anchoring-and-adjustment, also called anchoring effect, means an individual will tend to access or estimate the target value using a starting value when lacking information and as a result, the decisions are influenced and biased toward the reference point [33]. Social proof, also called information social influence, refers to the tendency of individuals to adopt the behaviors of surrounding people, assuming that others have more knowledge about the situation. The more uncertainty the individual perceives when assessing the situation, the more likely the individual will be biased toward others' opinions by employing social proof [34].

Messenger effect means an individual will tend to assign more weight or trust to information offered by an authoritative and knowledgeable entity regardless of the actual information quality [35]. Thus, the decision will be biased toward the opinion delivered from the entity considered as credible or authoritative.

Since the 1990s, people began to employ nudging techniques based on the understanding of cognitive processing to guide better human decisions in a large-scale. The exploitation of nudging first became popular in public policy for the purpose of promoting public welfare and then extended to digital environments (e.g., online social media and shopping sites) to affect user behaviors. Recommendation badges affect user decisions by engaging users in automatic thinking, where cognitive biases can be used to explain how heuristic shortcuts are established [30]. Sometimes these heuristic shortcuts are caused by a single cognitive bias and sometimes they can be led by multiple biases. For example, both *Amazon's Choice* and *Best Seller* can prompt a consumer to buy a product through the anchoring effect when she reaches a certain cognitive load. Additionally, *Best Seller* can nudge the user through the effect of social proof, while *Amazon's Choice* can nudge the user through the messenger effect. The following subsection introduces literature on nudging in detail.

2.3. Nudge and transparency

Nudge refers to a choice architecture that alters an individual's behavior in a forecastable way while preserving all available options without significantly changing the economic incentives [36]. Nudges are small modifications in an environment that are inexpensive and simple to implement, and have been widely employed in healthcare, public policy, and business, and reinvented with AI/ML-based algorithmic components in the digital context [37,5]. Nudging techniques are categorized into different kinds based on the influence mechanism and the common types are the following: informing and reminding (e.g., health warnings on cigarette packages and advertisements[5]), defaults (e.g., opt-in vs. opt-out nudges in different domains such as organ donor status in driving licenses[6]), social proof heuristics (e.g., "Customers Who Bought This Item Also Bought" shown on e-commerce platform[7]), and salience of the desired option [38,39,40]. For example, increasing the salience of the desired option means individuals' attention is drawn toward a specific option and people will be more likely to select that choice.

Digital nudge refers to the use of user interface (UI) design elements to guide user's behavior and decision-making in digital choice environments [41,13]. Digital choice environments used by both individual consumers and business customers include websites, mobile apps, games, and software [42,43]. In the digital context, nudging has become a promising field in IS research and has gained increasing attention from scholars [12,43,41,13]. Unlike other approaches that influence decision-making, such as those used in persuasive technology (e.g., addictive use of Facebook and TikTok[8]) [42,18], digital nudging relies on heuristics and cognitive biases and aims to lead the individual to desired option while preserving his or her freedom of choice through simple changes in the digital choice architecture.

Prior studies have discussed the principles of nudging mainly from two perspectives: one is the implementation of nudging; the other is the ethics of nudging, which is our focus in this study. Thaler and Sunstein [36] highlighted the ethical implications as a key consideration when making nudges. They proposed a standard way of designing nudges in which one must understand what biases the audience is subject to in a given situation and how this bias affects decision-making. Hansen and Jespersen [44] suggested that nudges affect both behaviors dominated by *system 1* and *system 2* (i.e., making rational choices) while the latter is achieved by switching from reflective thinking to the automatic mode via cognitive heuristics. Beyond that, they distinguished nudges as being transparent or non-transparent. In their definition, transparency means that the intention behind the nudge and the mechanism of behavior manipulation caused by the intervention are transparent to the target audience. For example, the footprints painted on the floor or street that leads people to the stairs or trash bin.[9] Non-transparent nudge means the nudged agent is not aware of the intention behind the nudge or cannot explain the inner workings of behavior change. For instance, the removal of trays in cafeterias intends to decrease food waste [44].

However, the definition of transparency in traditional settings cannot catch up with nudging in digital settings as it fails to account for the dynamism brought by the algorithmic components in the digital environment, where the AI nudge can automatically produce some desired outcome and the mechanism of nudging occurrence is hidden in the algorithm's inner "black box" [17]. For example, following the definition of transparent/non-transparent nudges in prior literature, recommendation badges (e.g., Amazon's Choice, Our Pick Today, and Best Seller) are considered transparent because customers can easily grasp their intentions (i.e., recommend a certain product in the search results) and the behavior change can be explained by psychological effects (e.g., anchoring effect and social proof). However, the badge Amazon's Choice is inherently less transparent than Best Seller because its placement is drawn from AI-driven algorithm and the mechanism of deployment is unpredictable. For AI nudging, consumers can understand the design intention behind the badge and the desired behavior nudged by the badge (i.e., choose the promoted product), but cannot reconstruct the reasoning process of how the badge recommendations are generated. Building on prior literature, our study extends the concept of nudge transparency within digital and AI-algorithmic contexts and emphasizes the transparency in how AI delegator implements and deploys nudges. We argue that AI-algorithmic recommendation badges can help consumers alleviate cognitive overload during the product selection process. However, they may also introduce uncertainties into the decision-making process due to their "black box" characteristic and potentially lower the decision quality.

2.4. AI Explainability

Research in Explainable Artificial Intelligence (XAI) has discussed similar issues about transparency and reached the consensus that explainability is a necessity for fair, accountable, and trustworthy AI [45,46,9,47]. XAI refers to the system that can display an interpretable chain of reasoning from the user's command, through the system's knowledge and inference to the targeted behavior, and emphasizes the movement and efforts made in response to the trust concerns and transparency of AI [48,49]. Research in this field has explored different dimensions of AI explainability, including evaluating instruments, forms, objectives, stakeholders, and the quality [9,50,51]. Moreover, XAI is one of the key requirements for implementing responsible AI. To support the responsible adoption of AI, organizations must embed ethical principles into AI applications and processes by building AI

[5] Cigarette labeling and health warning requirements on FDA.gov. See https://www.fda.gov/tobacco-products/labeling-and-warning-statements-tobacco-products/cigarette-labeling-and-health-warning-requirements, accessed Apr 26, 2023.

[6] Organ donation card design employes nudge with defaults heuristics. See https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8128443/, accessed Apr 26, 2023.

[7] "Customers Who Bought This Item Also Bough" is a nudge based on social proof heuristics. See https://www.nudgify.com/social-proof/, accessed Apr 26, 2023.

[8] Examples of apps using persuasive technology. See https://www.humanetech.com/youth/persuasive-technology, accessed Apr 26, 2023

[9] Green Nudge: Nudging Litter into The Bin. https://inudgeyou.com/en/green-nudge-nudging-litter-into-the-bin/ Retrieved on May 21, 2023.

systems grounded in trust and transparency [52]. In this paper, we focus on users' perceived transparency of the mechanism of the AI-driven recommendation badge. Even though existing studies focused on developing XAI methods to interpret the AI prediction and understand the inner workings of AI, there is still a lack of understanding of the users' perceptions and cognitive processing in different contexts of human-AI interaction from a socio-technical perspective, such as AI nudging [8]. By studying the recommendation badge implemented by AI/ML-based algorithm in the online shopping context, this paper can bridge the gap between digital nudging literature and the XAI literature, and pave the way for more AI nudging research in IS.

3. Research model and hypotheses development

This study examines how consumers' decision-making outcomes are influenced by the transparency of recommendation badges in the online shopping context. We posit that users' perception of recommendation badge transparency (*AI nudge* vs. *non-AI nudge*) can affect their perception of decision quality (i.e., *choice confidence*) during product selection and this impact is moderated by recommendation dissonance (*preference mismatch* vs. *preference match)*. To empirically assess the propositions, this study developed the research model (see Fig. 1) and a set of testable hypotheses in the following paragraphs.

In the online shopping context, recommendation badges placed by either AI algorithms or rule-based algorithms (e.g., using certain criteria such as sales volume) are used to nudge customers toward buying the product nominated by the platform. First, the user screens all the products and aims to make a rational decision (i.e., find a product that best matches her preferences), where reflective thinking (i.e., System 2) is involved. During the selection, the consumer may notice the recommendation badges alongside the platform-recommended product, which aim to ease cognitive load and help consumers make decisions faster and easier. Therefore, the consumer's judgment can be affected by the recommendation badge as heuristic cues can intervene and trigger the domination of automatic thinking (i.e., System 1) in decision-making [27]. Moreover, such a nudging setting can influence reflective thinking by introducing cognitive biases (e.g., anchoring effects), leading the user to put greater weight on the prompted product when evaluating its attributes against her preferences [44].

The significant difference between AI nudge and non-AI nudge resides in the recommendation generation mechanism. According to the official description offered by Amazon.com, *Amazon's Choice* is a recommendation badge delegated by an AI/ML-based algorithm that aims to discover products that other customers frequently choose for similar shopping needs.[10] It highlights what customers tell matters most to them and changes constantly as customer preferences and feedback evolve. Such an AI nudge is neither transparent nor explainable in terms of how the recommendation is generated. In contrast, *Best Seller* is generated based on a transparent criterion-specifically, a real-time ranking determined by sales volume within the relevant product category-and its placement is traceable.[11] Although the platform claims that no brand or selling partner can sponsor to be badged as *Amazon's Choice* (a major cause of non-transparency in prior literature), users do not know the full set of factors driving the placement of the badge nor the subjective and objective factors that contribute to ratings and popularity on the platform [10].

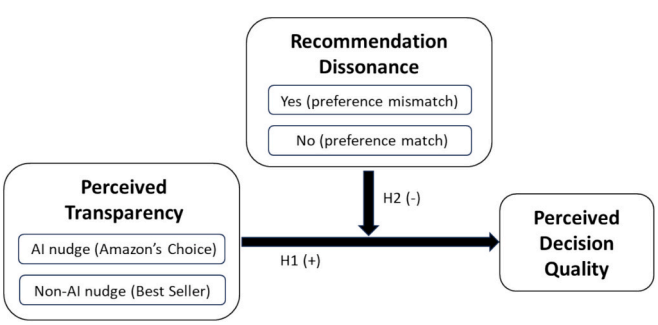

Fig. 1. Research model.

Drawing from dual-system theory and literature on cognitive bias, System 2 involves deliberate, logical, and effortful reasoning, which allows individuals to critically evaluate their decisions with available information and evidence. This process often leads to greater subjective decision quality (i.e., confidence) because decisions are based on thorough analysis rather than intuitive judgment based on System 1, which will foster a deep understanding of the decision-making process [53]. In contrast, System 1 relies on fast and intuitive judgments that are susceptible to cognitive biases. While System 1 is efficient for routine or low-stakes decisions, it lacks the clarity and rigor of System 2's reasoning, which is grounded in objective evidence. As a result, decision-making relying on System 1 may be associated with low subjective decision quality. For example, prior study showed that reflective capacity, individual's ability and tendency to engage in reflective thinking (i.e., System 2), is linked to increased self-confidence and better performance in the educational settings [54]. Nudging recommendations with low process transparency (e.g., AI nudge) provide little to no information about the underlying mechanism. As a result, they bring greater uncertainty into cognitive thinking process during product selection, causing consumers more likely to engage in System 1, which may subsequently reduce their subjective decision quality [55,56].

From another perspective, drawing from prior literature on transparency and knowledge-based trust, transparency is considered essential in designing trustworthy technology and fostering user trust [57,58,59]. Studies have found that transparency is positively associated with perceived integrity and can help improve fairness and reduce biases in e-commerce marketplaces [60,10]. Hence, the uncertainty and missing information about the inner workings of the recommendation badge may acquire less trust from the users and lower their choice confidence [31,10,34]. Accordingly, we posit Hypothesis 1 on the main effect of perceived transparency on user's choice confidence in the decision:

H1. Perceived transparency of the nudging recommendation in the choice architecture is positively related to the choice confidence of the user decision.

If the recommendation does not align with the users' product preferences, it increases uncertainty within the choice architecture as users become aware of the contradictory recommendation. In such case, users' choice confidence tends to further decrease because the recommendation dissonance will amplify the uncertainty caused by the lack of transparency in the underlying mechanism of nudging recommendation (i.e., AI nudge) and the decision-making process is even more likely to be dominated by System 1 [55,61]. While in the preference match condition, users will experience less uncertainty and doubt because the product recommendation aligns with the optimal product derived from their reflective thinking (System 2). Users are likely to perceive higher decision quality when they can clearly understand the underlying mechanism behind badge placement (e.g., non-AI nudge), especially when the recommended product aligns with their own assessment of the optimal choice. Hence, the high uncertainty caused by both low transparency and preference mismatch can lead to even lower choice

[10] FAQ page of Amazon's Choice. See https://www.amazon.com/b?ie=UTF8&node=21449952011, accessed Feb 16, 2023.

[11] Amazon's ranking list of bestselling items are well organized in each category and updated frequently. See https://www.amazon.com/Best-Sellers/zgbs, accessed Apr 28, 2023

confidence and vice versa.

From another perspective, drawing on the literature on knowledge-based trust, when customers cognitively perceive a misalignment between their preferences and the recommendation, they tend to form a lower assessment of the platform's capability to produce high-quality search results and have lower trust in the platform [62,63]. As a result, users will have lower choice confidence in decisions made within such a choice architecture. Accordingly, we propose the following hypothesis:

H2. The effect of perceived transparency on the user's choice confidence in the decision will be contingent upon recommendation dissonance (preference match vs. preference mismatch). Such an effect will be stronger for the preference match condition.

4. Research methodology

Two pretests were conducted to explore the online shopping interface and the difficulty of product choices in the online shopping environment. An online experiment was conducted to test the hypothesized relationships. The experiment controlled the product homogeneity (i.e., each product has no difference except the differentiating features) and presented participants with randomly assigned product orders (six power banks), badge placement (*Amazon's Choice* and *Best Seller*), and shopping sites (Amazon.com and Wish.com). The experiment controlled the effect of platform (e.g., large & famous platform vs. small & unknown platform) by displaying consistent user interface elements except for the platform logo. For example, in the Wish.com setting, we adopted both AI nudge and non-AI nudge in badge recommendation (i.e., *Wish's Choice* vs. *Best Seller*). While in the Amazon setting, we used these two badges: *Amazon's Choice* and *Best Seller*.

4.1. Experimental design

This study employs a 2 (type of badge: AI nudge and non-AI nudge) x 2 (recommendation: preference mismatch and preference match) between-subject factorial design. There are four scenarios in each of the two experiment settings (i.e., Amazon.com and Wish.com): 1. AI nudge and preference mismatch; 2. AI nudge and preference match; 3. non-AI nudge and preference mismatch; 4. non-AI nudge and preference match. Each scenario shared the same script that every participant was invited to pick a product as a gift for her friend based on the detailed specifications and explanations. After which, she was asked to fill out a questionnaire.

4.1.1. Design of websites and product placements

This study referred to the top-down list of search results from Amazon.com and adapted the interface to the online experiment scenarios to simulate the real-world online purchase experience. The website interface (Fig. 2) is inserted after the experiment scenario description as a high-resolution picture and presented to the participants before the survey. The experiment selected power bank as the product and each participant was displayed with a one-page product list from the search result. We selected the product category of power bank to ensure the randomization, realism and control of different scenarios in the experiment. First, its price is moderate, and prices across different power bank products show minimal variation, making a uniform price in our scenarios realistic. Second, the number of product features on a power bank is manageable, allowing for better control over participants' product preferences in the preference match/mismatch scenarios. Third, because many similar power banks are sold by different brands, participants are less likely to be biased toward a specific brand. All the product information and user interface elements are captured from Amazon.com and are reconsolidated into a webpage. This simulative setting not only makes participants feel as if they are shopping on e-commerce website, but also ensures the search results are consistent and not influenced by each participant's past browsing history.

In the product list, each item belongs to one brand and this study picked six noteless brands from the large population (over 1000 power banks) sold on Amazon.com. Even though online platforms (i.e., Best-Buy, Amazon, Apple Store, Steam, etc.) offer much more alternatives from a greater number of brands, existing experimental studies used smaller samples of alternatives for controlling the cognitive load of participants [23,26]. Moreover, humans can only process a limited amount of information concurrently and seven (plus or subtract two) is the number of items that can be held in a user's short-term memory [64]. Combining the results from two pretests and prior experiments about RA/RS, this study presents six products on one page and believes it is proper for controlling the participants' cognitive load [26,25].

The six products accompanied by the specifications are randomly placed on the page and the same order is shown to all users in the same website group. As shown in Table 1, this study created twelve website interfaces for the four scenario groups, and in each group, 3 variations of product lists are available for random assignment. Consumer preferences are affected by the product order and are susceptible to the influence of the shopping environment [65]. The randomization of products in each scenario group can help rule out the unobservable effects as well as the common method bias in the experiment procedures.

4.1.2. Design of product features and preference mismatch

In each scenario, two products can perfectly match the participant's preferences (referred to as *preferred* products) while the other four products cannot match the preferences (referred to as *unpreferred* products). Participants established their preferences based on the experiment description, which will be addressed in the following section of Experiment Tasks and Procedures. Detailed product features are shown in Table 3, where each product has nine common features and three differentiating features between *preferred* and *unpreferred* categories. As shown in Table 1 and Table 2, the first *preferred* product in each scenario is randomly placed in one of the first three positions (the 1st, 2nd, or 3rd position) and the second *preferred* product in each scenario is randomly placed in one of the last three positions (the 4th, 5th, or 6th position). The recommendation badge is assigned to the first *preferred* product in the preference match scenarios (scenario 2 and 4) and assigned to the first *unpreferred* product in the preference mismatch scenarios (scenario 1 and 3). As a result, the recommendation badge always appeared on the top placements in the search list, which simulates the real-world online shopping circumstance and ensures the nudge setting via the anchoring effect. In the design of the product features, the preferred products have better features and are more cost-effective than the unpreferred products given that these products are sold at the same price.

4.1.3. Instrument development

To ensure the reliability and validity of the measurement items, this study adapted existing instruments from prior studies and modeled all constructs as reflective, measuring with a seven-point Likert scale. The dependent variable to evaluate consumer decision quality is *choice confidence* [66]. We also captured experiment treatments and manipulation checks (see Table 4), including *recommendation dissonance* and *brand familiarity* [68].

4.2. Experiment task and procedures

All participants were randomly assigned to one of the fifteen website interfaces as well as one of the four experimental groups. They were told

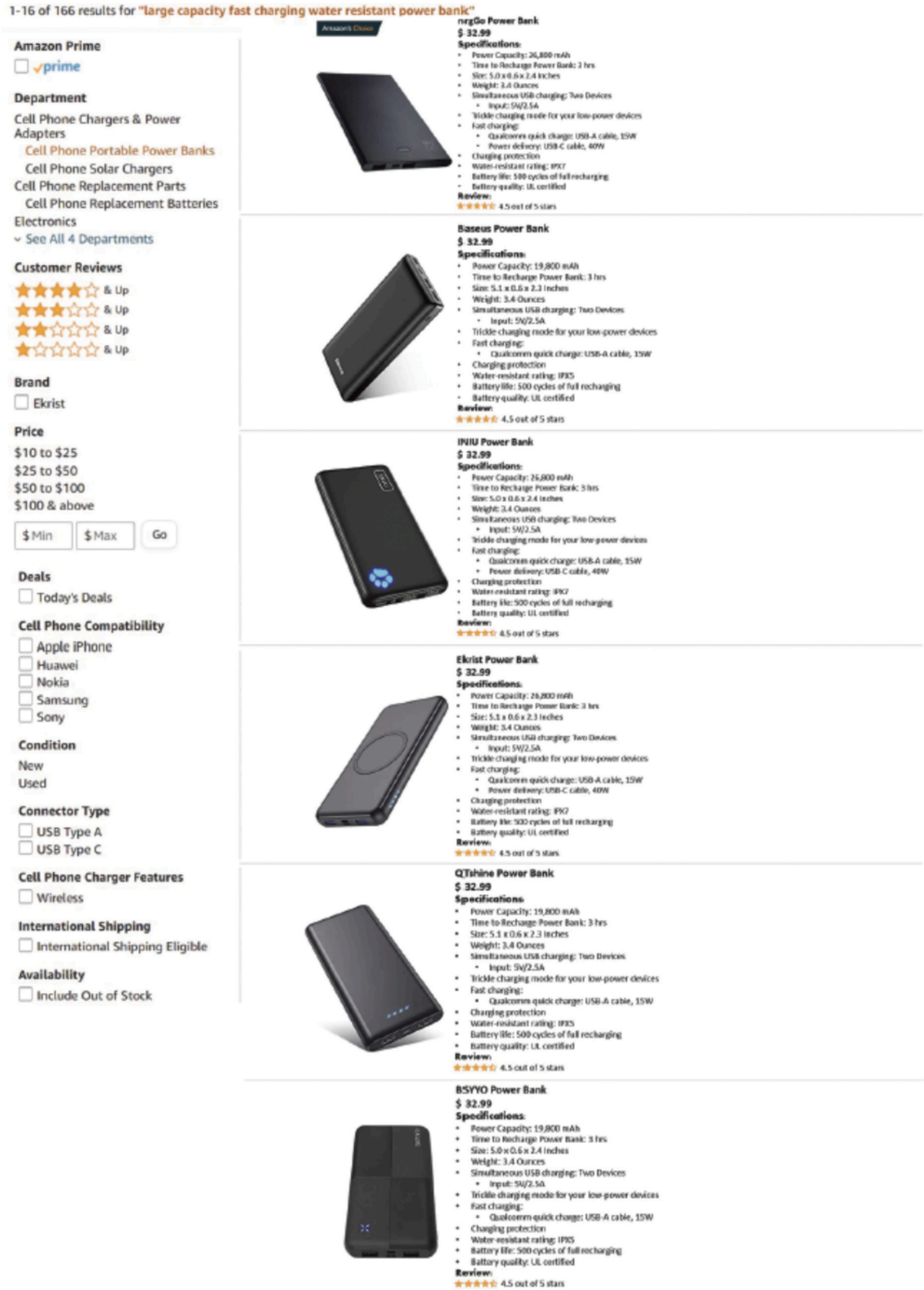

Fig. 2. Experiment webpage.

Table 1
Randomized product placements.

Website	Scenario	Condition	1st product	2nd product	3rd product	4th product	5th product	6th product
1	Scenario1	Amazon's Choice + Mismatch	BSYYO	Baseus	QTshine	nrgGo	Ekrist	INIU
2	Scenario1	Amazon's Choice + Mismatch	nrgGo	Baseus	INIU	Ekrist	QTshine	BSYYO
3	Scenario1	Amazon's Choice + Mismatch	QTshine	BSYYO	Ekrist	Baseus	nrgGo	INIU
4	Scenario2	Amazon's Choice + Match	nrgGo	Baseus	Ekrist	QTshine	INIU	BSYYO
5	Scenario2	Amazon's Choice + Match	QTshine	BSYYO	nrgGo	Ekrist	INIU	Baseus
6	Scenario2	Amazon's Choice + Match	BSYYO	QTshine	Ekrist	nrgGo	INIU	Baseus
7	Scenario3	Best Seller+ Mismatch	INIU	BSYYO	nrgGo	Baseus	Ekrist	QTshine
8	Scenario3	Best Seller+ Mismatch	Baseus	BSYYO	Ekrist	nrgGo	QTshine	INIU
9	Scenario3	Best Seller+ Mismatch	QTshine	Baseus	BSYYO	Ekrist	INIU	nrgGo
10	Scenario4	Best Seller+ Match	INIU	BSYYO	QTshine	Ekrist	Baseus	nrgGo
11	Scenario4	Best Seller+ Match	QTshine	INIU	Ekrist	Baseus	nrgGo	BSYYO
12	Scenario4	Best Seller+ Match	Baseus	nrgGo	QTshine	INIU	BSYYO	Ekrist

Table 2
Randomized preferred product and badge placements.

Website	Scenario	Condition	Preferred 1	Preferred 2	Badge
1	Scenario1	Amazon's Choice + Mismatch	2nd product	5th product	1st Product
2	Scenario1	Amazon's Choice + Mismatch	3rd Product	4th product	1st Product
3	Scenario1	Amazon's Choice + Mismatch	1st Product	4th product	2nd Product
4	Scenario2	Amazon's Choice + Match	3rd Product	5th product	3rd Product
5	Scenario2	Amazon's Choice + Match	3rd Product	6th product	3rd Product
6	Scenario2	Amazon's Choice + Match	1st Product	5th product	1st Product
7	Scenario3	Best Seller+ Mismatch	1st Product	4th product	2nd Product
8	Scenario3	Best Seller+ Mismatch	3rd Product	5th product	1st Product
9	Scenario3	Best Seller+ Mismatch	2nd Product	6th product	1st Product
10	Scenario4	Best Seller+ Match	2nd Product	4th product	2nd Product
11	Scenario4	Best Seller+ Match	1st Product	5th product	1st Product
12	Scenario4	Best Seller+ Match	3rd Product	4th product	3rd Product

Table 3
The design of product features.

Both preferred product and unpreferred product
Time to Recharge Power Bank: 3 h
Size: 5.0 × 0.6 × 2.4 Inches (slightly different for each one, e.g., 5.1 × 0.6 × 2.3 Inches)
Weight: 3.4 Ounces
Simultaneous USB charging: two devices
Input: 5 V/2.5 A
Trickle charging mode for low-power devices
Battery life: 500 cycles of full recharging
Review Ratings: 4.5
Charging protection
Preferred product
Power Capacity: 26,800 mAh
Water-resistant rating: IPX7
Fast charging: Qualcomm quick charge (USB-A cable, 15 W) & power delivery (USB-C cable, 40 W)
Unpreferred product
Power Capacity: 19,800 mAh
Water-resistant rating: IPX5
Fast charging: Qualcomm quick charge (USB-A cable, 15 W)

to buy a power bank on an e-commerce website as a gift for a friend who will use it mostly for camping trips. The script described the product requirements[12] in detail and emphasized the differentiating features, such as 25,000 mAh power capacity, waterproof, and fast power delivery. Moreover, the experiment script introduced *Amazon's Choice (Wish's Choice)* and *Best Seller* based on the information offered on Amazon.com (Wish.com) and clarified the differences between these two badges.

Participants were first required to read through the scenario script and tutorial about power bank features. Next, they were asked to open the captured webpage displaying the search result list. Upon completing the scanning of the product list and decision-making, they were asked to fill out a questionnaire that included attention checks, quiz questions to assess understanding of the task and the context, and measures of the dependent variables, perceptual constructs, and demographics of the subjects. Fig. 3 visualizes the experimental flow in the study.

4.3. Sample

Participants for this study were 837 college students in the business major recruited from a large major university in the U.S. and were provided with a credit-based incentive for their assistance in the study. The sample size ensured a 0.8 power for the required hypothesis tests to detect a medium effect size of 0.15. The invitation text of the experiment was posted on a recruitment website and broadcasted by the platform administrator via email to the subject pool.

4.4. Pretest

Two pretests were conducted before the main experiment to explore users' perceptions of badge transparency between AI nudge and non-AI nudge. The pretests also provided evidence that the manipulation of preference mismatch is reasonable (i.e., neither too hard nor too easy to detect the differences in product features among the six products). The pretests confirmed that the manipulations of product homogeneity (i.e., participants' use experience and brand familiarity among the products are averagely low and show no difference) and nudge effectiveness were successful and robust. Moreover, the pretests offered feedback for revising the phrasing of survey items and adapting the established

[12] The product requirements were designed to match the friend's needs of camping trips. The products available on the e-commerce webpage that best matched these requirements were the *preferred* products, not the *unpreferred* products.

Table 4
Instrument measurement.

Instrument	Measurement
Choice confidence	Measured subjectively with "choice confidence" (on a seven-point Likert scale from "strongly disagree" to "strongly agree"), adapted from Pereira [66]. • I am satisfied with the power bank that I chose for my friend from the list of items. • I am confident that, from the listed items, I selected the best power bank to suit my friend's needs.
Perception of badge transparency	Measured subjectively with "perceived transparency" (on a seven-point Likert scale from "strongly disagree" to "strongly agree"), adapted from Wang et al. [7] and Wang and Wang [10]. • The reasoning process of the recommendation badge is clear to me. • I could easily understand this recommendation badge's reasoning process. • This recommendation badge's logic in providing purchase recommendation was clear to me. • I understand the factors that will lead xxx (Amazon/Wish) to use the badge.
Perception of product homogeneity	Measured subjectively with "brand familiarity" (on a seven-point Likert scale from "strongly disagree" to "strongly agree"), adapted from Komiak and Benbasat [67]. • I am familiar with at least some of the brands listed in the search results. • I have bought at least some of the brands listed in the search results.
Web experience	Measured subjectively with "web experience" (self-reported options include "Never", "0–30 min", "30–60 min", "1–2 h", "2–3 h", and "More than 3 h"), adapted from Wang and Wang [10]. • On average, how much time per day do you spend on each of the following web activities: Accessing information on the web about products and services you may buy? • On average, how much time per day do you spend on each of the following web activities: Shopping (i.e., actually purchasing something) on the web?
Recommendation dissonance (manipulation check)	Captured objectively: Whether the recommendation badge is placed on the preferred product (preference mismatch vs. preference match).
Badge type (manipulation check)	Captured objectively: Whether the recommendation badge is AI nudge or non-AI nudge (Amazon's Choice or Best Seller).

Table 5
Demographic background of participants.

Variable/item		No. of participants (%)	Mean
Age		–	20.3
Gender	Male	375 (44.8 %)	–
	Female	462 (55.2 %)	–
Web Experience			
On average, how much time per day do you spend on each of the following web activities: Accessing information on the web about products and services you may buy?	Never	14 (1.7 %)	–
	0–30 min	299 (35.7 %)	–
	30–60 min	234 (28.0 %)	–
	1–2 h	145 (17.3 %)	–
	2–3 h	73 (8.7 %)	–
	More than 3 h	72 (8.6 %)	–
On average, how much time per day do you spend on each of the following web activities: Shopping (i.e., actually purchasing something) on the web?	Never	67 (8.0 %)	–
	0–30 min	397 (47.4 %)	–
	30–60 min	184 (22.0 %)	–
	1–2 h	106 (12.6 %)	–
	2–3 h	52 (6.2 %)	–
	More than 3 h	31 (3.7 %)	–

Table 6
Measurement model fit index.

Measure	Threshold	Measurement Model
$\chi 2$	–	55.054
DF	–	31
$\chi 2$/DF	Between 1 and 3	1.776
NNFI	>0.95	0.991
CFI	>0.95	0.994
SRMR	<0.08	0.028
RMSEA	<0.06	0.030

Table 7
Loadings and cross-loadings of measures.

	Perceived transparency	Choice confidence	Brand familiarity	Web experience
PerTran1	**0.851**	0.174	0.072	0.087
PerTran2	**0.903**	0.185	0.076	0.092
PerTran3	**0.876**	0.179	0.074	0.09
PerTran4	**0.744**	0.152	0.063	0.076
ChoConfi1	0.19	**0.928**	0.007	−0.014
ChoConfi2	0.176	**0.858**	0.007	−0.013
Brand1	0.069	0.006	**0.819**	0.156
Brand2	0.068	0.006	**0.809**	0.154
WebExp1	0.076	−0.012	0.141	**0.740**
WebExp2	0.082	−0.012	0.153	**0.803**

The bold in table highlights the loadings of the items of a construct on that construct.

measurement to this study.

In the first pilot study, each scenario contains five products and only one product has superior or inferior features compared to the other four products. Each product was attached with six common features and four differentiating features. The result showed that four differentiating features are too salient for the preference mismatch condition, which means less uncertainty in the thinking process. Moreover, it is practically less possible in the real-world situation. Hence, we decided to increase one more product and reduce the number of differentiating product features. In the second pilot study, each scenario contains six products, and half of the products have superior or inferior features compared to the other three products. Each product was listed with seven common features and one differentiating feature. The result showed participants could easily detect that half of the products were inferior and thus, the cognitive effort required to make the decision was too low for the recommendation badge to exert the nudging effect. Based on the pretests, the final experiment adopted six power banks in the product list and each power bank was listed with twelve common features and three differentiating features.

5. Data analyses and results

The analysis for this study involved two stages. First, the measurement model was evaluated using both exploratory and confirmatory factor analysis in IBM SPSS 25 software. Second, the research model was tested using the multi-group analysis with covariance-based SEM in IBM AMOS v24 software. We followed this two-stage approach because it has been used in recent studies to estimate interaction effects [69,70,71,72,73]. This section reports the results of the randomized experiment, including demographic data, measurement model, manipulation checks, and hypotheses testing.

5.1. Demographic data

Table 5 outlines the characteristics of participants in the experiment. More females participated in the study than males. The majority of participants are between 18 and 30 years old. More than 60 % of the participants screen product and service information for at least 30 min

Table 8
Reliability, average variance extracted and construct correlation matrix.

	Means (S.D.)	CR	Cronbach's Alpha	AVE†	1	2	3	4
1. Perceived Transparency	4.912 (1.338)	0.909	0.91	0.715	**0.845**			
2. Choice Confidence	5.701 (1.130)	0.888	0.89	0.798	0.206***	**0.893**		
3. Brand Familiarity	2.109 (1.442)	0.797	0.80	0.663	0.084*	0.008	**0.814**	
4. Web Experience	2.971 (1.112)	0.746	0.74	0.596	0.078	−0.016	0.190	**0.772**

Significance of correlations: * $p < 0.05$, *** $p < 0.001$.
† The diagonal is the value of the square root of AVE.

Table 9
Levene's test of manipulation checks.

Construct	Experimental groups	F (1, 835)	Sig.	Method
Perceived transparency	Non-AI vs. AI nudge	1.472	0.225	ANOVA
Brand familiarity	Non-AI vs. AI nudge	1.009	0.315	ANOVA
	Preference mismatch vs. match	0	0.997	ANOVA

Table 10
Results of multi-group analysis.

	Recommendation dissonance			
	Preference mismatch	Preference match	χ2	Δχ2(df = 1)
Perceived transparency–> Choice confidence	0.13***	0.29***	155.842	5.4**

Significance: ** $p < 0.05$, *** $p < 0.01$.

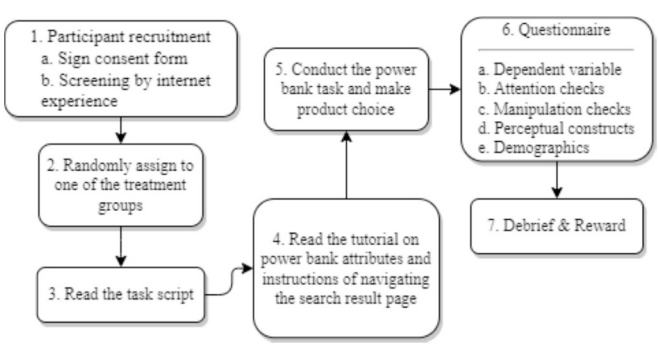

Fig. 3. Experimental flow.

each day. Moreover, over 44 % of the participants spent at least 30 min shopping online per day.

5.2. Measurement model

To assess measurement model fit, we followed the recommended fit indices in Hu and Bentler [74], including chi-square (χ2) with degree of freedom (DF), non-normed fit index (NNFI), comparative fit index (CFI), standardized root mean square residual (SRMR), and root mean square error of approximation (RMSEA). Table 6 describes each index and the corresponding threshold.

Confirmatory factor analysis, implemented in IBM SPSS Statistics 25, is used to assess the measurement properties of the dependent variable (except for objective measure) and other constructs (i.e., perceived transparency), all modeled as reflective constructs. We examined the reliability, internal consistency, and discriminant validity of each individual item based on the established guidelines [75,76]. All items showed good reliability for their corresponding constructs as the loadings of the measures exceeded 0.7 (see Table 7). The composite reliability and Cronbach's alpha of all the constructs exceed the threshold of 0.7 (Table 8), indicating good internal consistency of the measures. We also examined the convergent and discriminant validity following the guideline [75]. The results show that: (1) no item loads higher on a construct other than the one it measures (see Table 7); (2) the square root of Average Variance Extracted (AVE) of each of the two paired constructs is greater than the correlation of either construct in the pair (see Table 8).

5.3. Experiment manipulation checks

Manipulation checks were conducted for the experimental treatments to test the statistical differences of perceived transparency between the AI nudge and non-AI nudge. Following Hsu [77] and Stevens [78], we first conducted Levene's test between the two types of badges (Amazon's Choice and Best Seller) and then applied ANOVA or Welch method [79] correspondingly.[13] Based on the results of Levene's test (see Table 9), we conducted the manipulation checks accordingly [10].

Our data shows that participants' perceived transparency is significantly higher in the *non-AI nudge* condition than in the *AI nudge* condition (M = 5.078 vs. 4.738, F (1, 835) = 13.647, $p < 0.01$), suggesting that participants perceived higher transparency with non-AI nudge (e.g., Best Seller) than that with the AI nudge (e.g., Amazon's Choice).

We examined brand familiarity in different scenarios to address the concerns that consumers' familiarity and use experience of the product brand can affect the decisions (e.g., which product to choose). Our data analysis finds no significant difference in brand familiarity between *AI nudge* (e.g., Amazon's Choice) and *non-AI nudge* (e.g., Best Seller) conditions (M = 2.176 vs. 2.046, F (1, 835) = 1.710, $p = 0.191$) as well as no significant difference between *preference match* and *preference mismatch* conditions (M = 2.114 vs. 2.105, F (1, 835) = 0.008, $p = 0.927$), suggesting that there is no difference in consumers' familiarity and use experience of the product brands between different treatment groups.

We also assessed the self-reported awareness of the preference mismatch of the participants assigned to the "Amazon's Choice and preference mismatch" and "Best Seller and preference mismatch" conditions. The participants were asked to judge the following item in the post-task questionnaire: "I noticed that other products not recommended by the badge can satisfy all my needs better than the product recommended by the badge." We found that they were generally aware

[13] Levene's test is used to test homogeneity of variances. If null hypothesis is not rejected in Levene's test, then we use ANOVA test in the following analysis. If null hypothesis is rejected in Levene's test, then we use Welch method in the manipulation checks.

of the preference mismatch between the badge-recommended product and their purchasing needs (mean = 4.81, S.D. = 1.55). The result lent support to the preference mismatch manipulation in the badge recommendation.

5.4. Hypothesis testing

We first calculated the latent factor scores for each of the constructs using IBM SPSS Statistics 29 software and then ran the SEM modeling as well as multi-group analysis using IBM AMOS v24 software. The results of data analyses, including standardized path and path significance, are depicted in Fig. 4. We found that perceived transparency has a significant positive effect (path coefficient is 0.2, $p < 0.001$) on choice confidence of the purchase decision, supporting H1.

To test the moderation effect of recommendation dissonance on the relationship between perceived transparency and choice confidence, we conducted multi-group SEM (structure equation modeling) analysis, which can estimate separate slopes for each group of the categorical moderator and compare the changes in chi-squares between the unconstrained model and constrained model. This method has been widely used by prior studies when the moderator is categorical [80]. Our analysis showed that the restricted model exhibits a significant chi-square difference ($p < 0.05$) for the recommendation dissonance effect (see Table 10), lending support to the moderation effect of recommendation dissonance on the relationship between perceived transparency and choice confidence of the purchase decision. Moreover, the effect of perceived transparency is significantly stronger ($p < 0.001$) in the preference match group (path coefficient is 0.29) than in the preference mismatch group (path coefficient is 0.13). Hence, H2 is supported.

6. Discussion

The purpose of this study is to explore the impact of recommendation badges on users' decision quality in the context of online shopping. Drawing on the perspective of AI explainability, nudging, and transparency, we proposed a conceptual model including both the main effect of perceived transparency on choice confidence of the users' decisions and the moderation effect of recommendation dissonance on the main effect. Our empirical findings not only confirm the positive impact of perceived transparency on choice confidence but also indicate that for users facing recommendations matching their preferences, the direct impact is stronger.

6.1. Contributions to research

This study makes several theoretical contributions. First, this study contributes to the emerging body of research exploring the impact of AI-driven nudging on user decision-making in the context of online shopping. Despite the prevalence of recommendation badges based on AI algorithms in e-commerce platforms, there remains a gap in the empirical investigation of AI nudging in the literature. This finding represents an initial step toward exploring the impact of AI nudging on individual behavior, specifically through the lens of its lack of transparency. Since nudging can influence human behavior on a subconscious level and at scale, careful oversight is essential to monitor both the persuasive power and the unintended consequences of the black box nature of AI nudging-especially as AI nudging becomes more widely adopted. On one hand, this study presents evidence of the subtle nudging effect exerted by AI-powered recommendation badges on user decision quality. On the other hand, we call for more examinations of the side effects of outsourcing the nudge implementations to AI systems. A better understanding of the relationship between AI nudging and consumer decision will contribute to the nudging and IS literature and help online platforms realize potential ethical risks associated with algorithmically AI-driven nudges (e.g., undermining user decision quality).

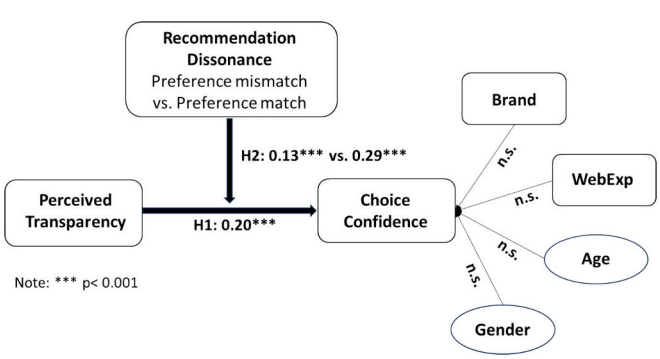

Fig. 4. SEM results.

Second, this study sheds light on the controversies surrounding algorithmic fairness and user trust in AI system. We demonstrated that AI nudges (e.g., Amazon's Choice and Wish's Choice) are perceived less transparent than non-AI nudges (e.g., Best Seller) due to their opaque AI-algorithmic mechanisms. The lack of transparency reduces users' ability to audit or verify recommendation logic with System 2 thinking and could enable potential biases (e.g., favoring Amazon-branded products). As a result, such opacity nature of the AI algorithm creates inequitable decision-making environments where users lack information parity with the e-commerce platforms. Our results highlight the necessity of standardized transparency metrics in platforms/systems that incorporate AI-algorithmic components, to ensure fair representation of recommendation criteria. We argue that fairness in AI systems requires transparency parity: users are supposed to have equivalent access to understanding recommendation logic as platforms have. Additionally, the results imply that consumers distrust systems when they cannot trace decision logic, even when the recommendation match their purchase preferences. The findings align with XAI principles that trust requires both outcome quality and process transparency.

Third, this study adds to the literature on recommendation agents/systems in IS by extending the online recommendation research to the digital nudging context (e.g., recommendation badges). Prior studies have indicated that biased personalized recommendations from product RA affect consumers' decision quality and decision effort [25]. This study conceptualized and empirically tested the moderation role of recommendation dissonance in the relationship between users' perceptions of transparency and their decision quality. This advancement considers the defects of personalized recommendations and examines the contingent effects in both preference match and preference mismatch (i.e., biased recommendation) conditions. The moderation results imply that online platforms should carefully inspect the deficiencies and biases (i.e., preference mismatch between recommendation and user) of the algorithms before delegating AI to construct the real-time choice architecture.

Last, the results of the study lend empirical support to the importance and necessity of AI-algorithmic transparentizing and calls for disclosure of the inner "black box" workings of the recommendation badges implemented by AI. Our findings imply the potential risk of large-scale manipulation of user decisions through AI nudging without platform disclosure (e.g., prioritizing proprietary products), underscoring the need for: 1) third-party auditing requirements, clear conflict-of-interest disclosures, and full-transparent AI-powered recommendations. Moreover, this study connects the nudging and IS literature by unveiling the impact of AI nudging on user behaviors. The key finding provides evidence that a lack of transparency of the badge placement mechanism weakens users' choice confidence during the decision-making processes. In the recommendation dissonance (i.e., preference mismatch) condition, the low-transparent recommendation badge (i.e., AI nudging) can lead to even lower choice confidence. The results of this study can also be generalized to other contexts where AI algorithms and nudging UI elements are combined to guide individual behaviors.

6.2. Implications for practice

A nudge always serves a specific purpose, and so does AI nudging. When humans delegate the role of the choice architect to the AI algorithm, the AI model may be proficient in nudging consumers to notice the product chosen by most other users who enter the same search keywords or helping users identify the product that perfectly matches their purchase preferences. However, it is also important and helpful to find out the other unintended and unwelcome side effects brought by the AI nudging other than the desired outcome. Once we understand the mechanism through which the AI nudge exerts the unwanted effect, we can work on improving or fixing the design.

This study provides three major implications for e-commerce platforms. First, it is critical for the practitioners to be aware of the side effects of AI nudging on user decision quality and the importance of badge transparency in online badge recommendation. Our findings demonstrate that user's subjective decision quality will be enhanced if platforms can disclose the inner workings of the recommendation badge (i.e., all the factors leading to the badge placement). Prior study presented similar findings [7], which mainly focuses on product sponsorship disclosure in the listing.

Second, since recommendation dissonance often occurs in online recommendations [26], practitioners should increase users' awareness of the preference mismatch through education initiatives about AI-powered recommendation mechanism as well as the commercial implications. When users possess this prior knowledge, they can adjust their expectations and judgments accordingly and may be more likely to engage in reflective thinking, which facilitates high-quality decisions. Moreover, our findings suggest platforms explore improved designs of recommendation badges supported by AI algorithms, aimed at debiasing in consumer decision-making. For instance, e-commerce platforms should adopt ethical design framework and pick recommendation badges with clearer language and a more straightforward explanation of how the recommendations are generated.

Third, the results of this study bring ethical implications to the use of AI/ML model. This study may motivate platforms (e.g., Amazon.com) to cherish their reputation and to be cautious about giving endorsements (e.g., Amazon's Choice) without revealing all the factors leading to the recommendation. This study also urges large-scale technology companies to follow the Federal Trade Commission (FTC) guidelines on disclosure and endorsement[14] and reveal key details- such as the full range of factors used in the AI algorithms-that drive the generation of recommendations in search results, especially nudging-based recommendations that harnesses underlying human cognitive processes.

6.3. Limitations and future research

As with other research, this study comes with certain limitations, and future developments are needed to resolve these unanswered questions. First, the data came from a single source, i.e., college students in business majors with online shopping experience assessed the dependent and independent variables in one post-task survey, which makes the finding subjective in nature. To address the concern of common method bias (CMB), we have taken several steps. Although the CMB is not a serious concern based on results from data analysis, future studies can improve on this limitation by using data collected from multiple sources, such as gathering secondary data on different platforms with objective measurements.

Second, the random experiment created an artificial environment for each participant to conduct a task and recorded the response with a survey instrument, which may not accurately represent the real-world conditions and limit the generalizability of the results. For example, participants' prior experience with the listed products in the search results and their familiarity with the power banks may influence decision quality in unobserved ways. Although our research hypotheses have received strong empirical support in a strictly controlled experiment, we acknowledge limitations in external validity and believe there are viable future opportunities to replicate and extend the current study in more realistic contexts.

Third, the participants and the e-commerce platforms described in the experiment are all US-based, which may raise the possibility of regional bias. Moreover, approximately 94 % of our participants were born after 1997 and the average age of them is 20. Although Gen Z (born between 1997 and 2012) represents about 20 % of the US population as of 2023,[15] scholars should be careful when interpreting the impact of AI nudging on consumer decision quality with the majority of target users in different age ranges. Additionally, the participants in this study are all college students, which may not fully represent the broad consumer behavior in the U.S. Although 53 % of U.S. population aged 25 and older hold postsecondary degrees, according to the U.S. Census Bureau (2022),[16] caution is still warranted when interpreting the results for consumers with different educational backgrounds. Future studies should consider including a more diverse group of participants-encompassing a wider age range and varying levels of education-to enhance the generalizability of the findings.

Lastly, this study focuses on the impact of AI nudging transparency on user decision quality. On one hand, it is also possible that such an impact can affect the user's actual product choice in the purchase decision. Future studies are encouraged to explore the impact of AI nudging transparency on other decision-making outcomes or individual perceptions (e.g., decision effort, extent of product search, risk tolerance, trust, etc.) that may contribute additional insights to research in digital nudging and AI. On the other hand, the underlying mechanisms that influence user qualities warrant further investigation in the context AI nudging. Specifically, future research should explore the impact of additional heterogeneous factors on user decision quality-such as AI aversion, social norms, product types, and contextual factors on user decision quality.

7. Conclusion

Grounded in the theoretical perspective of cognitive processing, nudging, and online recommendation research in IS, this study constructs and tests a research model that examines how consumers' subjective decision quality is influenced by their perceptions of transparency of recommendation badges on e-commerce platforms. The results of this study have not only confirmed the positive impact of perceived transparency on users' choice confidence but also tested the moderating effect of recommendation dissonance on the above main impact, where the preference mismatch condition is shown to undermine the positive influence of perceived transparency on choice confidence. With enriched understanding of user decision outcomes with recommendations based on AI nudging in the online shopping context, the findings of this study fill a void in the digital nudging literature, bridge a gap between nudging and AI literature, and respond to the recent call for interdisciplinary oversight of AI nudging in the IS literature.

[14] FTC guidelines for disclosure: https://www.ftc.gov/business-guidance/advertising-marketing/endorsements-influencers-reviews;

FTC guidelines for endorsement: https://www.ftc.gov/business-guidance/resources/ftcs-endorsement-guides-what-people-are-asking

[15] US population by generation: https://www.statista.com/statistics/797321/us-population-by-generation. Retrieved on Sep 1, 2024.

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CRediT authorship contribution statement

Yuxiao Luo: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Nanda Kumar:** Validation, Supervision, Project administration, Methodology, Conceptualization. **Adel Yazdanmehr:** Supervision, Conceptualization.

Declaration of competing interest

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Data availability

Data will be made available on request.

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